SMA Practical Codes

1. Study various SMA tools

1. Go to https://communalytic.org/
2. Sign In (clg account) and create New Project then Add dataset then add Youtube API key AIzaSyAfFegeZ_ALsLcpHN5IWwMkXXaRCSeyLLM
3. Paste any yt video url link and done

2. Data Collection for Social Media Platform

```
from yt_dlp import YoutubeDL
import pandas as pd
with YoutubeDL({"quiet": True}) as y:
   r = y.extract_info("ytsearch20:Laptop Review", download=False)
data = [[v["title"], v["uploader"], v["view_count"],
       v["like_count"], v["comment_count"],
       v["upload_date"], v["duration"], v["webpage_url"]]
      for v in r["entries"]]
df = pd.DataFrame(data, columns=[
   "Title","Channel","Views","Likes",
   "Comments","Upload Date","Duration","URL"
])
df.to_csv("youtube_business_data.csv", index=False)
print(df.head())
```

3. Data Cleaning & Storage of Social Media Data

```
from cleantext import clean
import pandas as pd
from pymongo import MongoClient
df = pd.DataFrame({
   "user":["john_doe","alice_99","mike_123","sara_k"],
   "tweet":[
      "I love AI!!! #MachineLearning http://example.com",
      "Worst service ever!!! #badexperience",
      "Python is awesome :) #coding #AI",
      "Delivery was late and bad #fail"
   ],
   "location":["Mumbai","Delhi",None,"Pune"]
})
df["cleaned_tweet"] = df["tweet"].apply(clean)   # fixed (no wrong params)
df["location"] = df["location"].fillna("Unknown")
col = MongoClient("mongodb://localhost:27017/")["social_media_db"]["cleaned_posts"]
col.delete_many({})
col.insert_many(df[["user","cleaned_tweet","location"]].to_dict("records"))
print(df)
```

4. Exploratory Data Analysis with visualization

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
# Simplest Data Dictionary
data = {
```

```
    "Channel": ["TechHub", "GadgetBox", "ReviewPro", "ByteSize", "TechHub"],
    "Views": [10000, 50000, 25000, 5000, 15000],
    "Likes": [500, 2000, 1200, 100, 800],
    "Comments": [50, 150, 80, 10, 60]
}
df = pd.DataFrame(data)
# Calculate Metric
df['Engagement %'] = (df['Likes'] + df['Comments']) / df['Views'] * 100
# Fast Plotting
plt.figure(figsize=(10, 4))
# Plot A: Engagement Rate by Channel
plt.subplot(1, 2, 1)
sns.barplot(x="Engagement %", y="Channel", data=df, estimator="mean")
# Plot B: Correlations
plt.subplot(1, 2, 2)
sns.heatmap(df.select_dtypes('number').corr(), annot=True, cbar=False)
plt.tight_layout(); plt.show()
```

5. Content based social media analysis

```
import pandas as pd, matplotlib.pyplot as plt
from textblob import TextBlob
from wordcloud import WordCloud
from collections import Counter
# Simple Data
data = ["I love this laptop! 😍", "Terrible battery life 😡", "Normal build quality", "Amazing speed! 🔥", "Bad screen"]
df = pd.DataFrame(data, columns=["Content"])
# Shortest Sentiment & Keyword Logic
df["Pol"] = df["Content"].apply(lambda t: TextBlob(t).sentiment.polarity)
df["Sent"] = df["Pol"].apply(lambda p: "Pos" if p > 0 else ("Neg" if p < 0 else "Neu"))
words = " ".join(data).lower().split()
# Print Summary
print(df[["Content", "Sent"]], "\nTop Words:", Counter(words).most_common(3))
# Quick Visuals
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(10, 4))
df["Sent"].value_counts().plot(kind="pie", ax=ax1, title="Sentiment")
ax2.imshow(WordCloud(background_color="white").generate(" ".join(words)))
ax2.axis("off")
plt.show()
```

6. Structure based Social Media Analysis Model

```
import pandas as pd, networkx as nx, matplotlib.pyplot as plt, community as louvain
# Data & Graph
df = pd.DataFrame({'s':['A','A','B','B','C','C','D','E','F','G','H','D','B'],
                   't':['B','C','C','D','D','E','E','F','G','H','A','G','H']})
G = nx.from_pandas_edgelist(df, 's', 't')
# Analysis
part = louvain.best_partition(G)
rank = nx.pagerank(G)
# Quick Visual & Output
nx.draw(G, with_labels=True, node_color=list(part.values()), cmap='Set3')
```

```
plt.show()
print("Communities:", part)
print("Top User:", max(rank, key=rank.get))
```

7. Dashboard & reporting tool on real tie social media data

https://github.com/Atharva12072004/X-Social-Media-Data-Analysis

run .pbix file in Power BI

8. Design Creative Content for promotion of Your Business On Social Media

Go to Canva and create one post for promotion

9. Analyse competitor activities using Social Media Data

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
# Competitor Data (Brand vs Competitor A & B)
data = {
   'Brand': ['You', 'Comp_A', 'Comp_B'],
   'Followers': [10000, 50000, 15000],
   'Total_Engage': [1200, 2000, 1800] # Likes + Comments
}
df = pd.DataFrame(data)
# Competitor Metric: Engagement Rate (True Influence)
df['Eng_Rate %'] = (df['Total_Engage'] / df['Followers']) * 100
# Quick Visual Battle
sns.barplot(x='Brand', y='Eng_Rate %', data=df, palette='magma')
plt.title("Who is Winning the Audience?")
plt.show()
print(df.sort_values('Eng_Rate %', ascending=False))
```

10. Social Media text analysis model

```
import pandas as pd, matplotlib.pyplot as plt
from textblob import TextBlob
from collections import Counter
# Simple Data (Posts)
posts = ["I love this product", "Bad service", "Great price!", "Very slow delivery", "Amazing
experience"]
# Analyze: Sentiment (Positive/Negative) & Word Count
sentiment = ["Positive" if TextBlob(p).polarity > 0 else "Negative" for p in posts]
words = Counter(" ".join(posts).lower().split()).most_common(3)
# Quick Visual & Output
pd.Series(sentiment).value_counts().plot(kind='bar', color=['green', 'red'])
plt.title("Text Sentiment Analysis"); plt.show()
print(f"Sentiment: {sentiment}\nTop Topics: {words}")
```